# MoveOn Phone Cover Downloader & Translation Pipeline

**Specification Version:** 1.2 | **Project:** inventory/coverstore | **Status:** Production

## 1. System Architecture & Scope

Automates single-product asset acquisition from **moveon.global**. Strictly executes against only the user-provided product URL with zero out-of-scope spidering. Auto-identifies phone cover materials, prompts for phone model selection, generates standardized SKU codes, replaces non-English text on photos with clean English translations, and creates a deduplicated e-commerce catalog in \cataloge\.

## 2. Custom SKU Generation Standard

All variation cover images are named using the standardized format:
**[model abbreviation]-[material abbreviation]-[color abbreviation]-[print abbreviation]**
*Example:* RN8-SIL-WT-S.jpg (Realme Neo8, Liquid Silicone, Antique White, Solid Color)

| Component | Code Format | Representative Examples |
|---|---|---|
| Phone Model | 2-6 Alphanumeric | `RN8 (Realme Neo8), IP15PM (iPhone 15 Pro Max), S24U (Galaxy S24 Ultra)` |
| Material | 3 Letters | `SIL (Liquid Silicone), GLS (Glass), TPU (TPU), LTH (Leather), PC (Hard PC)` |
| Color | 2 Letters | `WT (White), BK (Black), PR (Purple), GN (Green), BL (Blue), CL (Clear)` |
| Print / Finish | 1 Letter | `S (Solid/Cube), P (Printed), G (Gradient), E (Electroplated/Mirror), C (Clear)` |

## 3. Storage Hierarchy & Cataloge Deduplication

| Directory Path | File Naming & Contents |
|---|---|
| `\[model]\[material]\original\` | original_01.jpg ... Raw product showcase & gallery images. |
| `\[model]\[material]\sku\` | [SKU].jpg (e.g. RN8-SIL-WT-S.jpg) ... Variant covers. |
| `\[model]\[material]\translated\` | translated_[SKU].jpg ... English-translated graphics. |
| `\[model]\[material]\cataloge\` | **Curated Catalog:** Clean original photos (no text) + translated images, with all duplicates removed. |
| `\cataloge\` | Global catalog directory aggregating all deduplicated store images. |
| `\scripts\` | downloader.py, catalog_builder.py, sku_generator.py, moveon_client.py. |